Building and evaluation of a PBPK model for COMPOUND in healthy adults

Version	**master-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/COMPOUND-Model/releases/tag/vmaster
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

Table of Contents

1 Introduction

Dabigatran is an oral direct thrombin inhibitor administered as a pharmacologically inactive prodrug, dabigatran etexilate. After absorption, it undergoes enzymatic conversion via carboxylesterases (CES1 in the liver and CES2 in the intestine) to the active moiety dabigatran. Dabigatran shows low oral bioavailability (~6–7%), moderate inter-individual variability, and is predominantly eliminated unchanged via renal excretion.

In addition to renal clearance, dabigatran undergoes glucuronidation, forming dabigatran acylglucuronide, which contributes to total active exposure. Therefore, often the sum of dabigatran + dabigatran glucuronide = "total dabigatran" is quantified and reported. Absorption of the prodrug dabigatran etexilate is strongly influenced by P-glycoprotein (P-gp), making transporter activity a key determinant of systemic dabigatran exposure and drug–drug interaction potential.

The presented whole-body PBPK model is based on the model developed by Moj et al. 2019 and includes dabigatran, its prodrug dabigatran etexilate and the pharmacologically active metabolite dabigatran acylglucuronide. It was established to be used for DDI predictions as dabigatran etexilate is a widely used Pgp substrate in clinical DDI studies.

The herein presented model building and evaluation report evaluates the performance of the PBPK model for dabigatran in healthy adults.

The presented dabigatran PBPK model as well as the respective evaluation plan and evaluation report are provided open-source (https://github.com/Open-Systems-Pharmacology/Dabigatran-Model). The DDI performance is documented in the DDI qualification report (https://github.com/Open-Systems-Pharmacology/COMPOUND-Model).

2 Methods

2.1 Modeling Strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Regarding the relevant anthropometric (height, weight) and physiological parameters (e.g. blood flows, organ volumes, binding protein concentrations, hematocrit, cardiac output) in adults was gathered from the literature and has been previously published (PK-Sim Ontogeny Database Version 7.3). The information was incorporated into PK-Sim® and was used as default values for the simulations in adults.

The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available PK-Sim® Ontogeny Database Version 7.3 (Schlender 2016) or otherwise referenced for the specific process.

The PBPK model was developed using a stepwise modeling strategy. First, an intravenous dabigatran model was established to describe distribution and elimination processes, including UGT2B15-mediated glucuronidation of dabigatran to dabigatran glucuronide as well as the renal clearance of both.

As many clinical reports provide "total dabigatran" = dabigatran + dabigatran glucuronide plasma concentrations instead of dabigatran glucuronide measurements, an observer for "dabigatran sum" was incorporated and used in parameter identifications. In addition, observers for fraction excreted to urine following intravenous administration of dabigatran (for dabigatran glucuronide) and fraction excreted to urine following oral administration of dabigatran etexilate (for dabigatran and dabigatran glucuronide) were implemented.

Second, the model was extended to include the orally administered prodrug dabigatran etexilate. This step required incorporation of Pgp-mediated transport affecting its absorption and CES1/CES2-mediated prodrug activation to produce dabigatran.

The respective contributions of CES1 and CES2 were informed by data from the RE-LY trial (Paré 2013), reporting that patients carrying 2 minor CES1 alleles (modeled as complete loss of CES1 activity) show a 28% reduction of their total dabigatran plasma concentration trough values (dabigatran + dabigatran glucuronide).

While the original model considered data of 2 different clinical DDI studies (rifampicin-mediated Pgp induction (Härtter 2012) and clarithromycin-mediated Pgp inhibition (Delavenne 2013), for the model update data of 7 further clinical DDI trials were used, adding Pgp inhibition by rifampicin (n=2), itraconazole (n=1), verapamil (n=1), and additional clarithromycin studies (n=3) (please see the DDI qualification report (https://github.com/Open-Systems-Pharmacology/COMPOUND-Model).

Unknown parameters were identified using the PK-Sim® Parameter Identification module. Model evaluation included comparison of predicted versus observed concentration–time profiles, goodness-of-fit plots, and geometric mean fold error (GMFE) analysis.

Details about input data (physicochemical, *in vitro* and clinical) can be found in Section 2.2.

Details about the structural model and its parameters can be found in Section 2.3.

2.2 Data

2.2.1 In vitro / physico-chemical Data

A literature search was performed to collect available information on physicochemical properties of dapagliflozin. The obtained information from literature is summarized in the table below.

Dabigatran etexilate

Parameter	Unit	Value	Source	Description
MW	g/mol	627.73	Moj 2019	Molecular weight
pK_a		4.0, 6.7	Zhao 2014	Acid dissociation constant
Solubility (pH)	mg/mL	1.8 (7.0)	Moj 2019	Aqueous Solubility
logP		3.8	Zhao 2014	Partition coefficient between octanol and water
fu	%	7	Moj 2019	Fraction unbound in plasma

Dabigatran

Parameter	Unit	Value	Source	Description
MW	g/mol	471.51	Moj 2019	Molecular weight
pK_a		4.1, 4.4, 12.4	Zhao 2014	Acid dissociation constant
Solubility (pH)	mg/mL	0.017 (7.0)	Moj 2019	Aqueous Solubility
logP		-2.2	Zhao 2014	Partition coefficient between octanol and water
fu	%	65	Moj 2019	Fraction unbound in plasma
B/P ratio		0.67	Moj 2019	Blood to plasma ratio

Dabigatran glucuronide

Parameter	Unit	Value	Source	Description
MW	g/mol	647.23	Moj 2019	Molecular weight
pK_a		4.1, 4.4, 12.4	Moj 2019	Acid dissociation constant
logP		-4.15	Moj 2019 (in-silico)	Partition coefficient between octanol and water
fu	%	65	Moj 2019	Fraction unbound in plasma

2.2.2 Clinical Data

A literature search was performed to collect available clinical data on dabigatran in healthy adults.

2.2.2.1 Model Building

The following studies were used for model building (training data):

Source	Arm / Treatment / Information used for model building
1160-0005 internal clinical trial report 2001	Intravenous infusion of 0.1 mg (single dose)
1160-0005 internal clinical trial report 2001	Intravenous infusion of 1.0 mg (single dose)
1160-0005 internal clinical trial report 2001	Intravenous infusion of 5.0 mg (single dose)
1160-0061 internal clinical trial report 2006	Oral capsule administration of 110 mg and 150 mg (twice daily)
1160-0194 internal clinical trial report 2014	Oral solution administration of 150 mg (twice daily)
1160-0194 internal clinical trial report 2014	Oral capsule administration of 150 mg (twice daily)
Blech 2008	Oral solution administration of 200 mg (single dose)
Härtter 2012	Oral capsule administration of 150 mg (single dose)

2.2.2.2 Model Verification

The following studies were used for model verification:

Source	Arm / Treatment / Information used for model building
1160-0074 internal clinical trial report 2009	Oral capsule administration of 150 mg (single dose)
1160-0082 internal clinical trial report 2008	Oral capsule administration of 150 mg (single dose)
Delavenne 2013	Oral capsule administration of 300 mg (single dose)

2.3 Model Parameters and Assumptions

2.3.1 Absorption

Absorption of dabigatran etexilate is influenced by intestinal permeability and Pgp transport. For the model parameter `Specific intestinal permeability` the calculated value was used. Parameter optimization of Pgp kcat was informed by addition of clinical Rifampicin-Dabigatran Pgp-induction DDI data to the parameter optimization.

The dissolution of capsules was implemented via an empirical Weibull function using the Weibull parameters from the original model by Moj et al. 2019.

2.3.2 Distribution

After testing the available organ-plasma partition coefficient and cell permeability calculation methods built in PK-Sim, observed clinical data was best described by choosing the partition coefficient calculation by `Rodgers and Rowland` and cellular permeability calculation by `PK-Sim Standard` for all three compounds. For the model parameter `Specific organ permeability` the calculated values were used for all three compounds, with the `Lipophilicity` values optimized to best match clinical data (see Section 2.3.1).

2.3.3 Metabolism and Elimination

- For dabigatran etexilate the final model includes transport by Pgp, prodrug ester cleavage by CES1 and CES2 and passive renal filtration (trace amounts excreted in urine following oral administration).
- For dabigatran the final model includes glucuronidation by UGT2B15 and renal excretion (70.6-76.2% excreted in urine following intravenous, 4.3% excreted in urine following oral administration).
- For dabigatran glucuronide the final model includes renal excretion only, implemented with the GFR fraction parameter values from the original model by Moj et al. 2019. Dabigatran glucuronide in urine is not meaningful in clinical pratice (2.7-6.4% excreted in urine following intravenous, 0.4% excreted in urine following oral administration) and we had very little data to inform its estimation.

2.3.4 Automated Parameter Identification

These are the results of sequential the final parameter identifications.

1. iv:

Model Parameter	Optimized Value	Unit
`Dabigatran logP`	1.144	
`Dabigatran UGT2B15 kcat`	4.299	1/min
`Dabigatran GFR fraction`	1.228	
`Dabigatran glucuronide logP`	-0.80	
`Dabigatran glucuronide GFR fraction`	6.36 (fixed)	

2. oral:

Model Parameter	Optimized Value	Unit
`Dabigatran etexilate logP`	1.809	
`Dabigatran etexilate Pgp kcat`	0.46	1/min
`Dabigatran etexilate CES1 kcat`	10.815	1/min
`Dabigatran etexilate CES2 kcat`	0.721	1/min
`Dabigatran etexilate GFR fraction`	1.00 (fixed)	

3 Results and Discussion

The PBPK model for dabigatran was developed and verified using clinical pharmacokinetic data from studies listed in Section 2.2.2. Furthermore, 9 different Pgp-DDI studies were used to qualify the model for its application as Pgp substrate for DDI predictions (please see the DDI qualification report (https://github.com/Open-Systems-Pharmacology/COMPOUND-Model).

The model was evaluated covering data from studies including in particular

- intravenous (infusion) and oral administration (solution and capsules, single and multiple dose)
- a dose range of 0.1 to 300 mg.

The model includes Pgp transport and CES1/CES2-mediated hydrolysation of the prodrug dabigatran etexilate, UGT2B15-mediated glucuronidation of dabigatran, and renal secretion of dabigatran glucuronide.

The next sections show:

1. the final model parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

3.1 Final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: DabiEtex

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.8 mg/ml		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	1.8088312812 Log Units	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32	Measurement	True
Fraction unbound (plasma, reference value)	0.07		Measurement	True
Is small molecule	Yes			
Molecular weight	627.73 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Transport Protein: ABCB1-FIT

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	0.08 µmol/l	
kcat	0.46 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Metabolizing Enzyme: CES1-FIT

Molecule: CES1

Metabolite: Dabigatran

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	24.9 µmol/l	
kcat	10.8149677921 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Metabolizing Enzyme: CES2-FIT

Molecule: CES2

Metabolite: Dabigatran

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	5.5 µmol/l	
kcat	0.7209978528 1/min	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-05 10:32

Systemic Process: Glomerular Filtration-DabiEtex

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Compound: DabiGluc

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.8 mg/ml		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	-0.8 Log Units	Parameter Identification-Parameter Identification-Value updated from '#11 - P-gp 0.46, DabiGluc logP -0.80, CESlink' on 2026-03-04 18:22	Measurement	True
Fraction unbound (plasma, reference value)	0.65		Measurement	True
Is small molecule	Yes			
Molecular weight	647.23 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-DabiGluc

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	6.36	

Compound: Dabigatran

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	17 mg/l		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	1.1436481967 Log Units	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56	Measurement	True
Fraction unbound (plasma, reference value)	0.65		Measurement	True
Is small molecule	Yes			
Molecular weight	471.51 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Dabi

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: UGT2B15-FIT

Molecule: UGT2B15

Metabolite: DabiGluc

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	512 µmol/l	
kcat	4.2993761606 1/min	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56

Formulation: DabiEtex_Capsule

Type: Weibull

Parameters

Name	Value	Value Origin
Dissolution time (50% dissolved)	1.0579 min	
Lag time	0 h	
Dissolution shape	0.2628	
Use as suspension	No	

3.2 Diagnostic Plots

Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data used presented in Section 2.2.2.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-1: GMFE for Goodness of fit plot for concentration in plasma

Group	GMFE
Intravenous infusion	1.16
Oral capsule	1.35
Oral capsule - Verification dataset	1.83
Oral solution	1.92
All	1.50

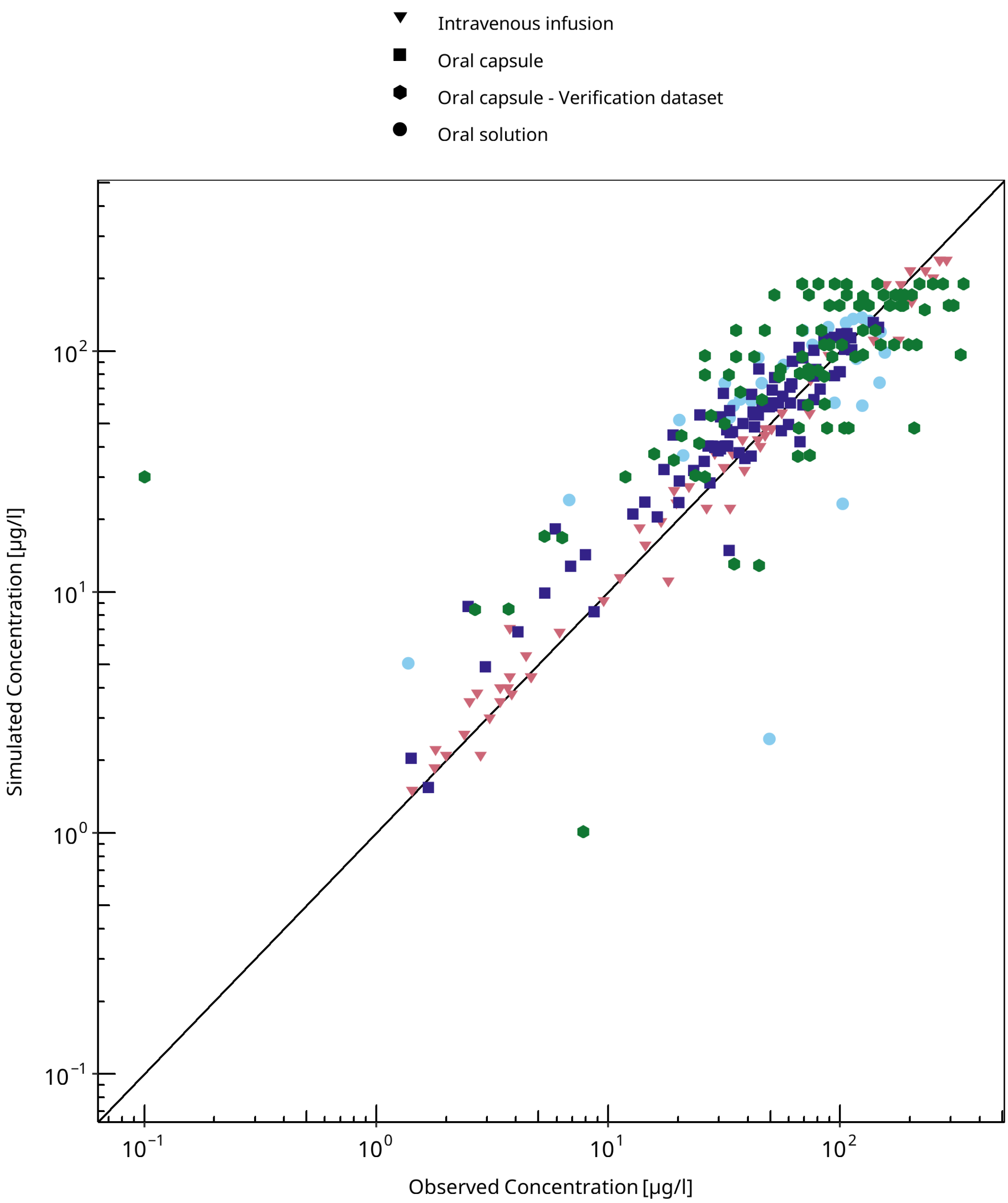

Figure 3-1: Goodness of fit plot for concentration in plasma

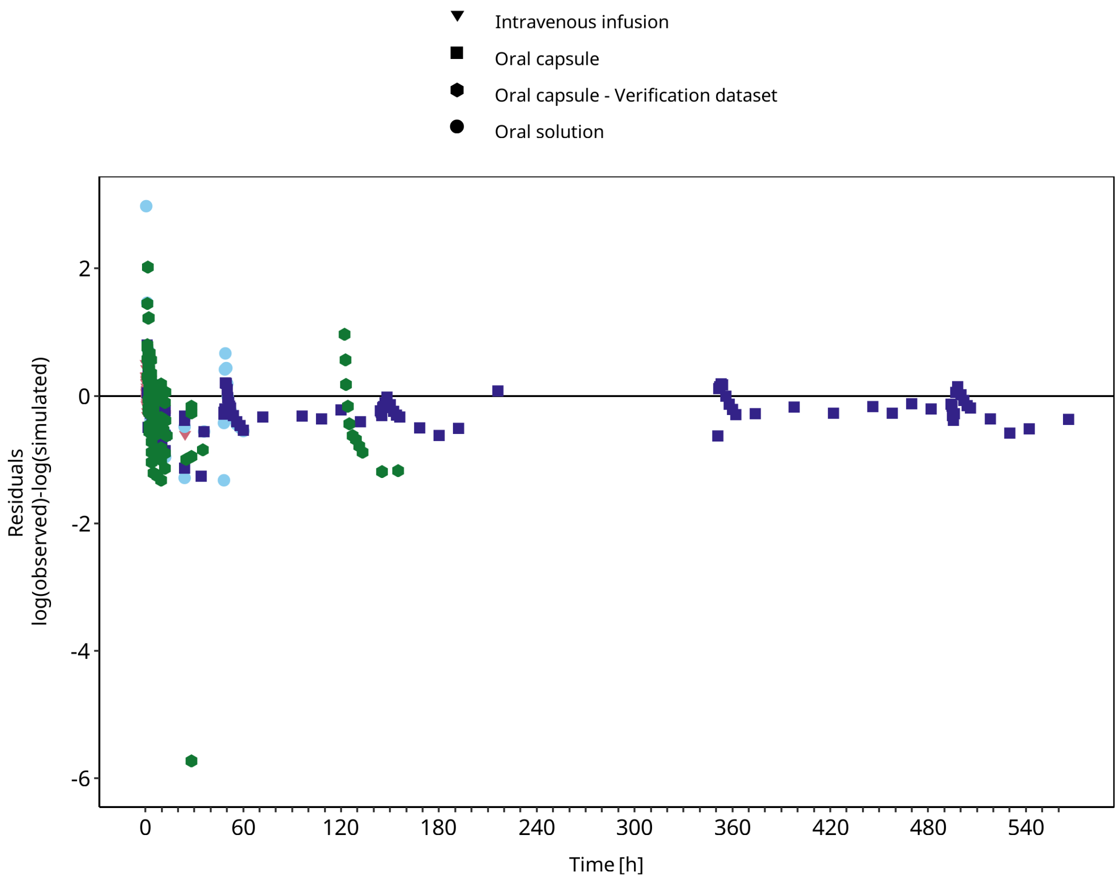

Figure 3-2: Goodness of fit plot for concentration in plasma

3.3 Concentration-Time Profiles

Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 are presented below.

3.3.1 Model Building

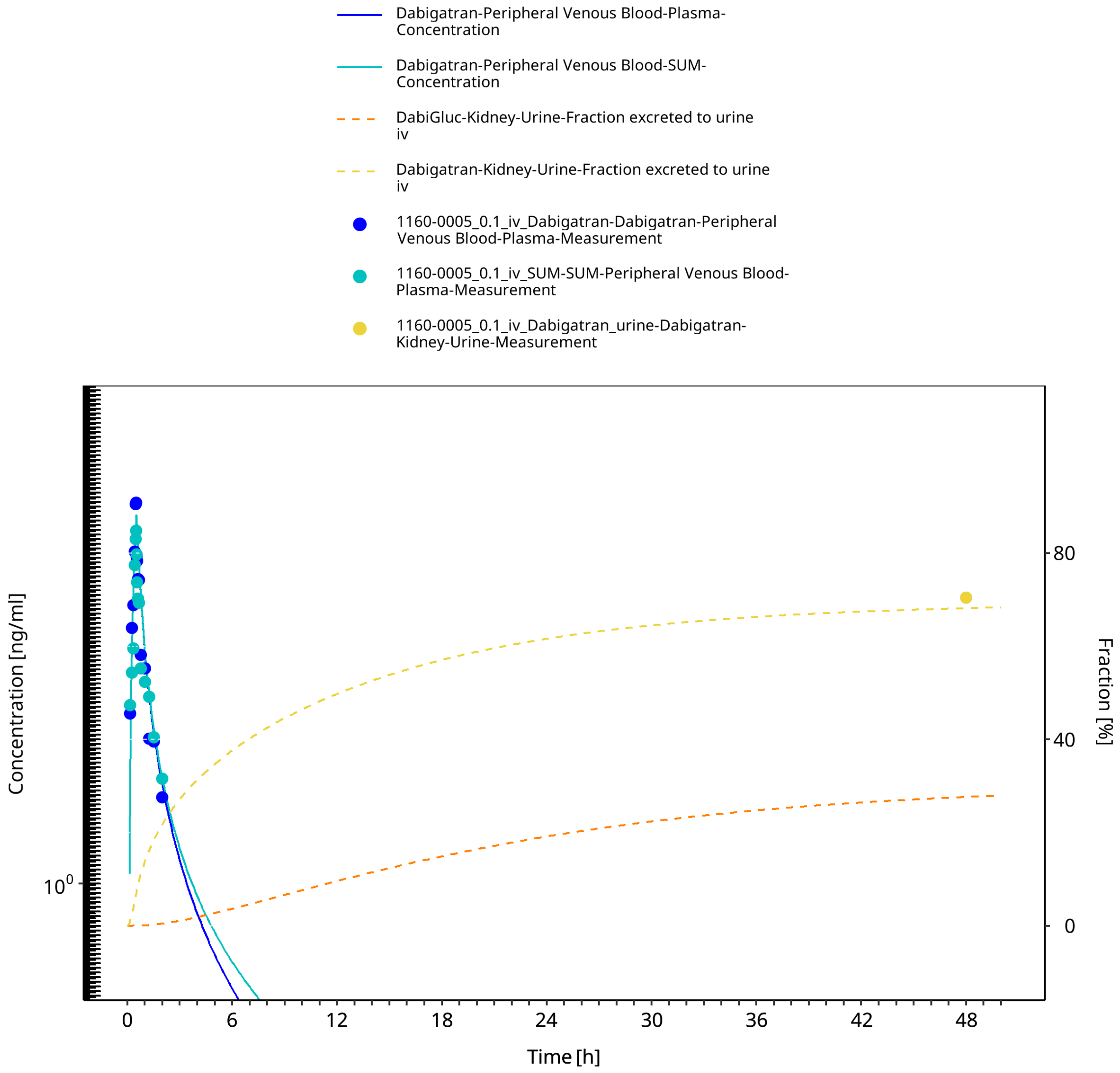

Figure 3-3: 1160-0005 - Dabigatran iv 0.1 mg - log

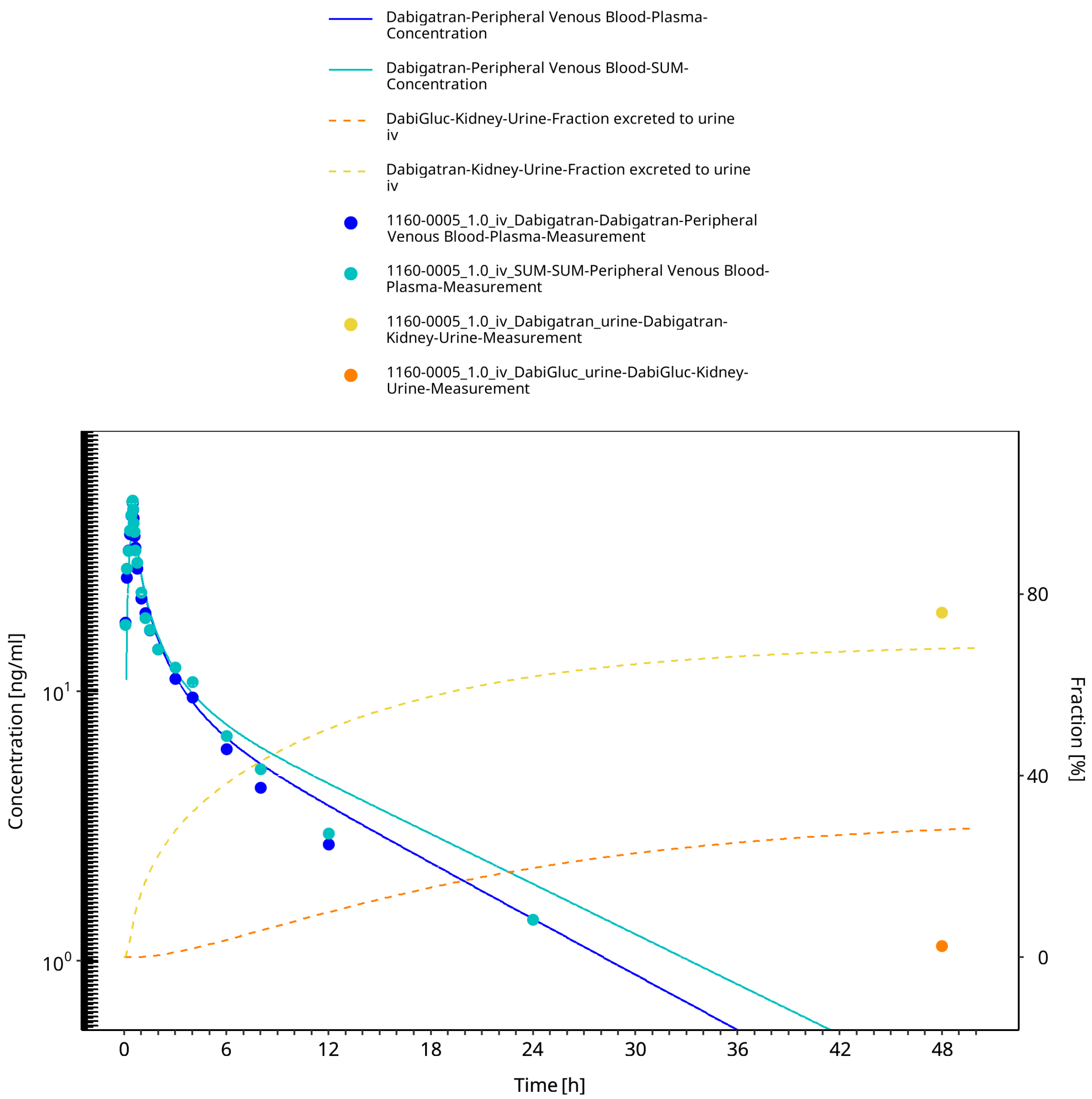

Figure 3-4: 1160-0005 - Dabigatran iv 1.0 mg - log

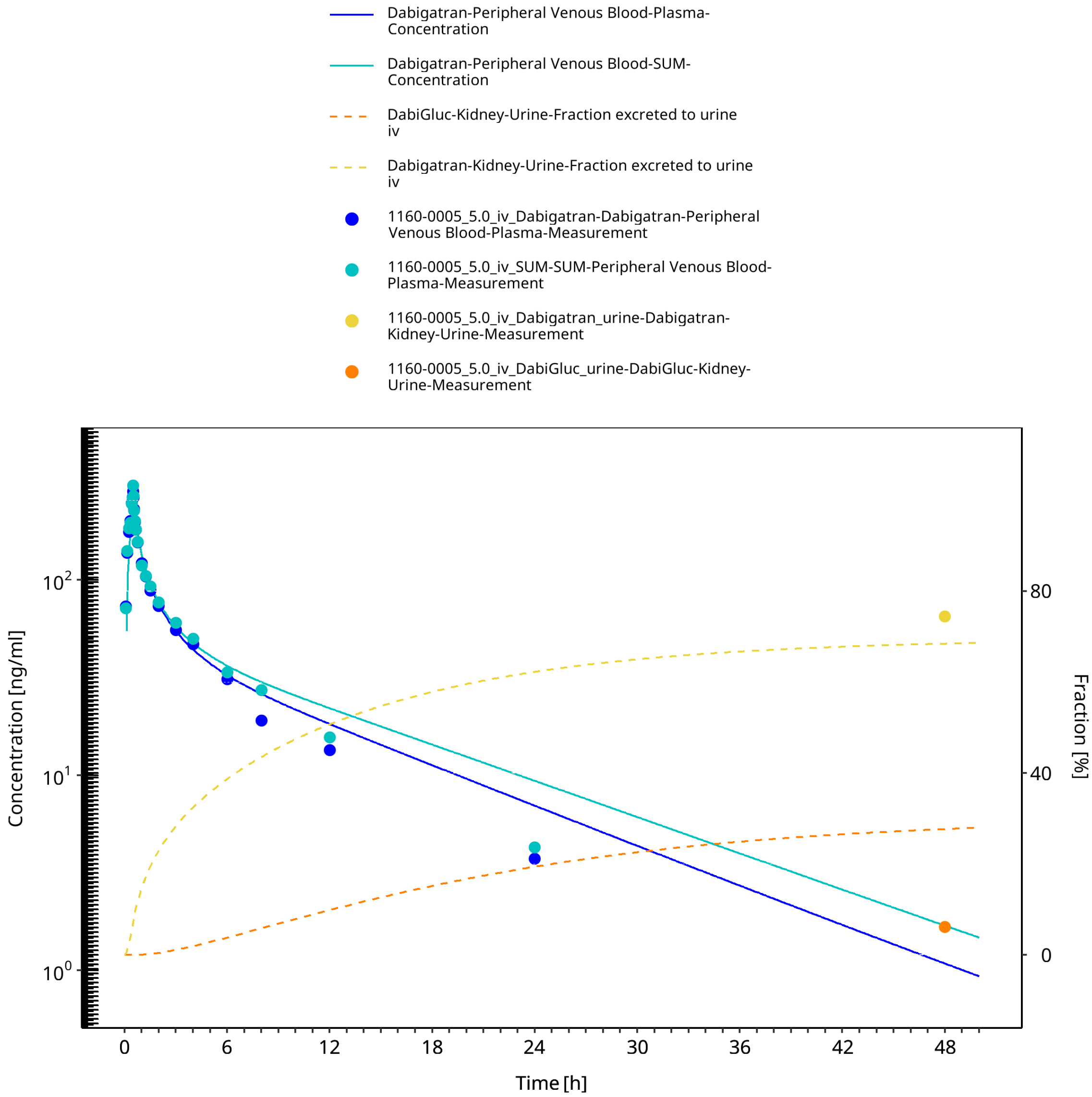

Figure 3-5: 1160-0005 - Dabigatran iv 5.0 mg - log

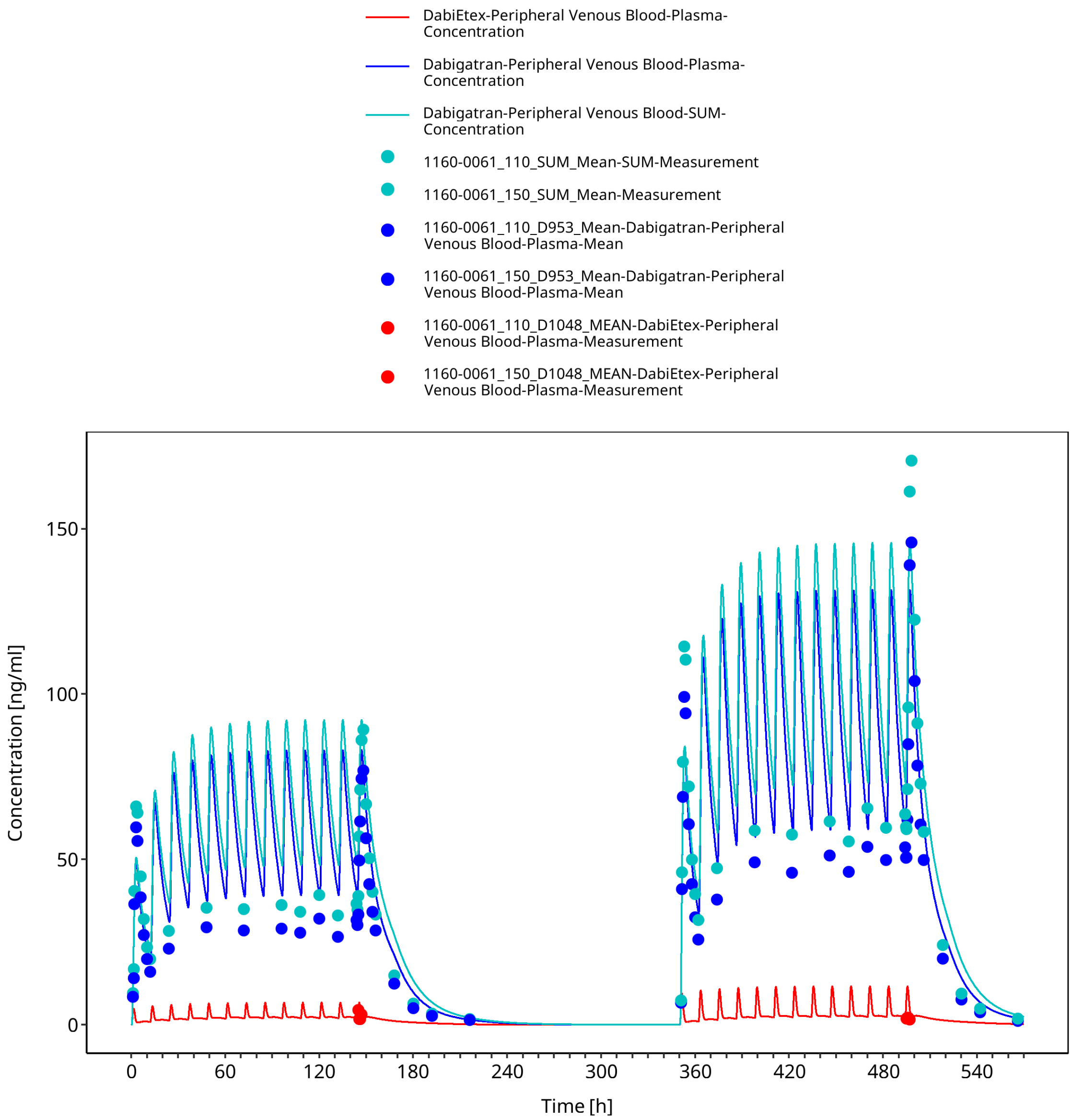

Figure 3-6: 1160-0061 - Dabigatran 110+150 mg BID - lin

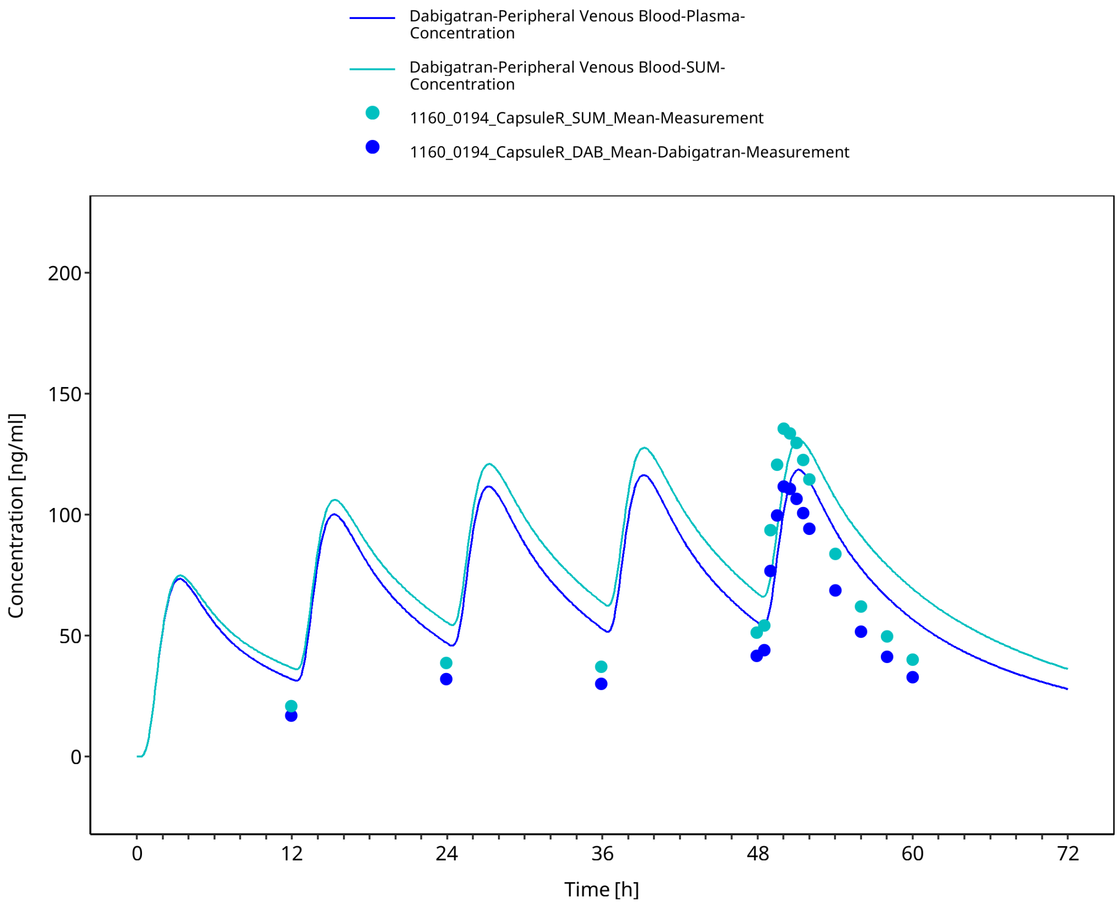

Figure 3-7: 1160-0194 - Dabigatran 150 mg Capsule - lin

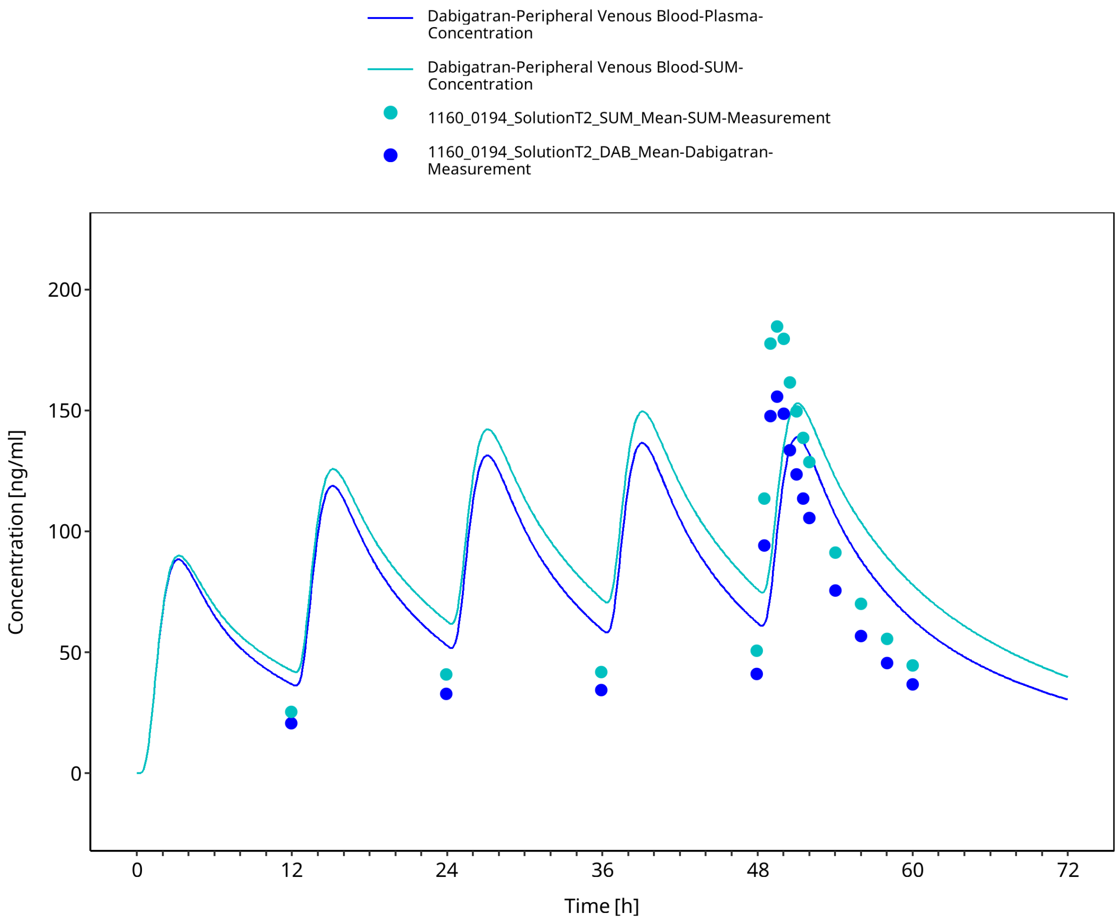

Figure 3-8: 1160-0194 - Dabigatran 150 mg Solution - lin

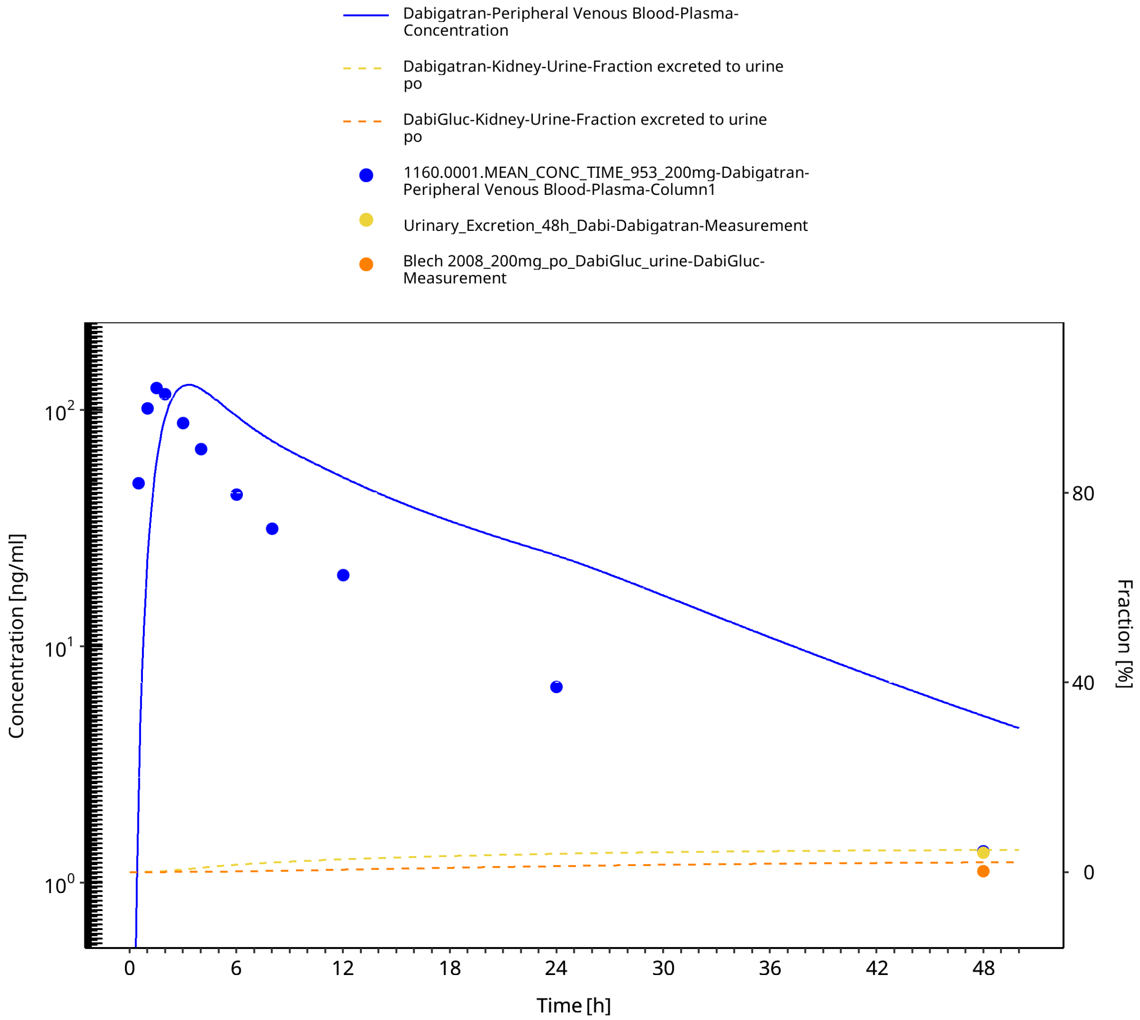

Figure 3-9: Blech 2008 - Dabigatran 200 mg Solution - log

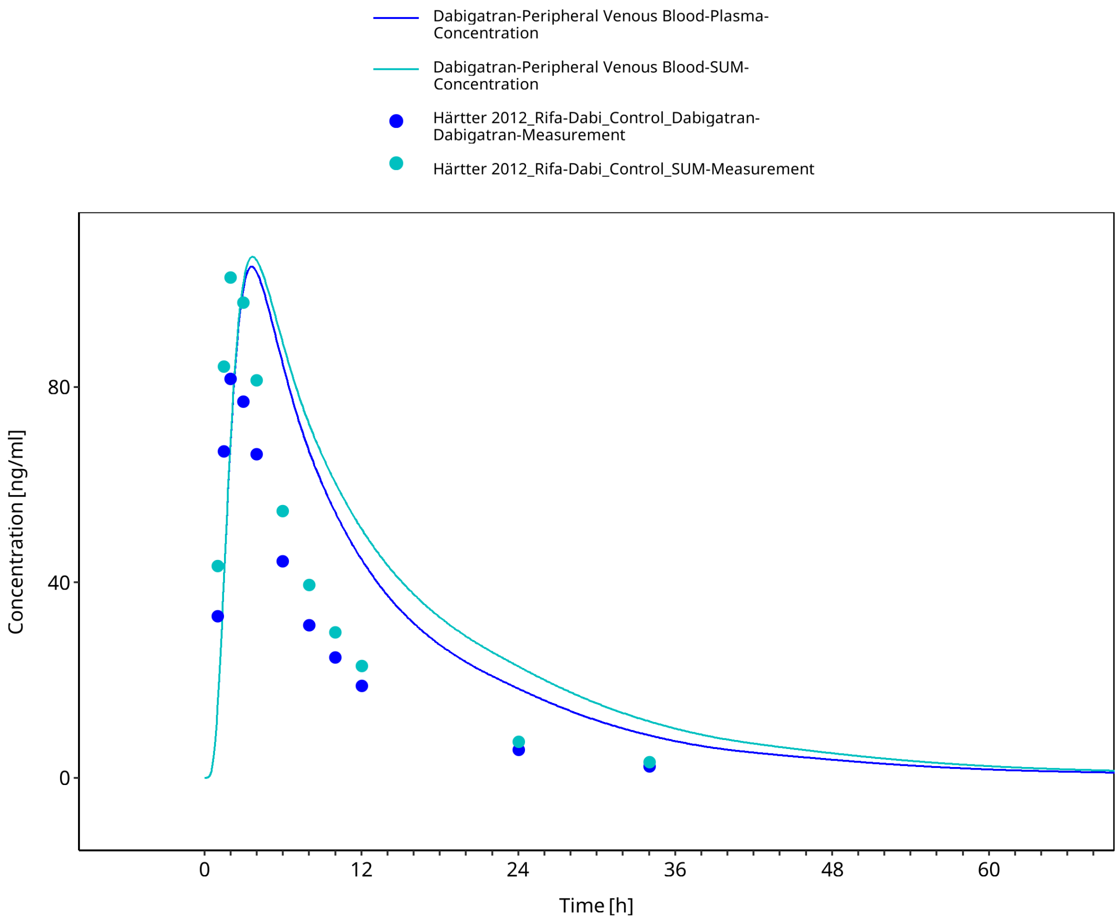

Figure 3-10: Härtter 2012 - Dabigatran DDI Rifampicin - lin

3.3.2 Model Verification

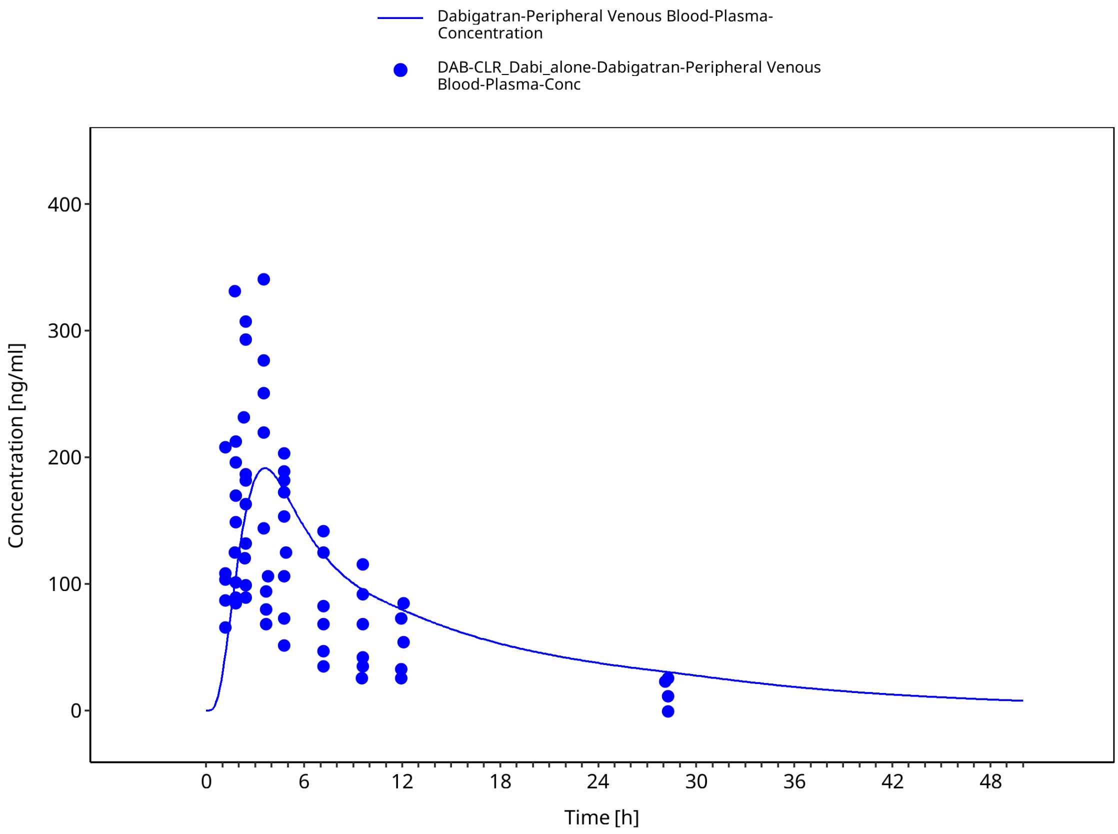

Figure 3-11: Delavenne 2012 - Dabigatran DDI Control - lin

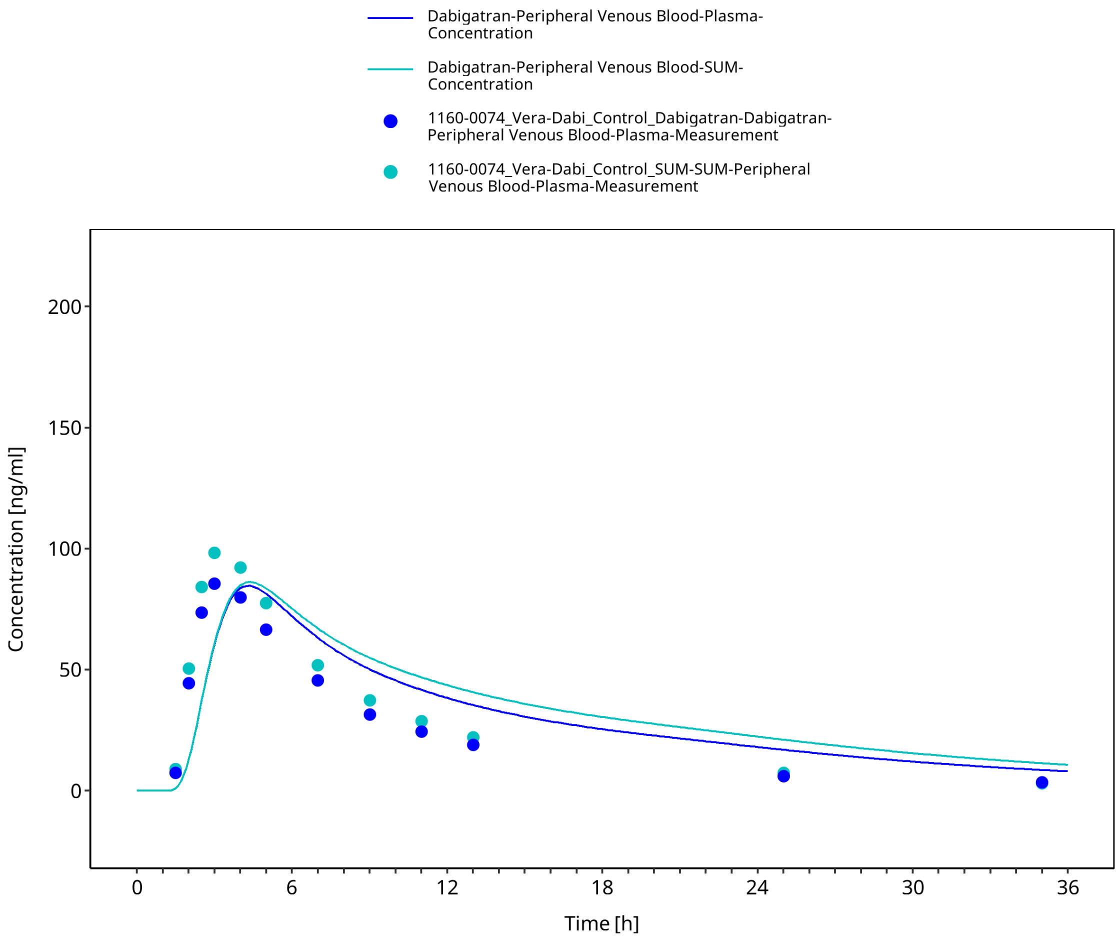

Figure 3-12: 1160-0074 - Dabigatran DDI Control - lin

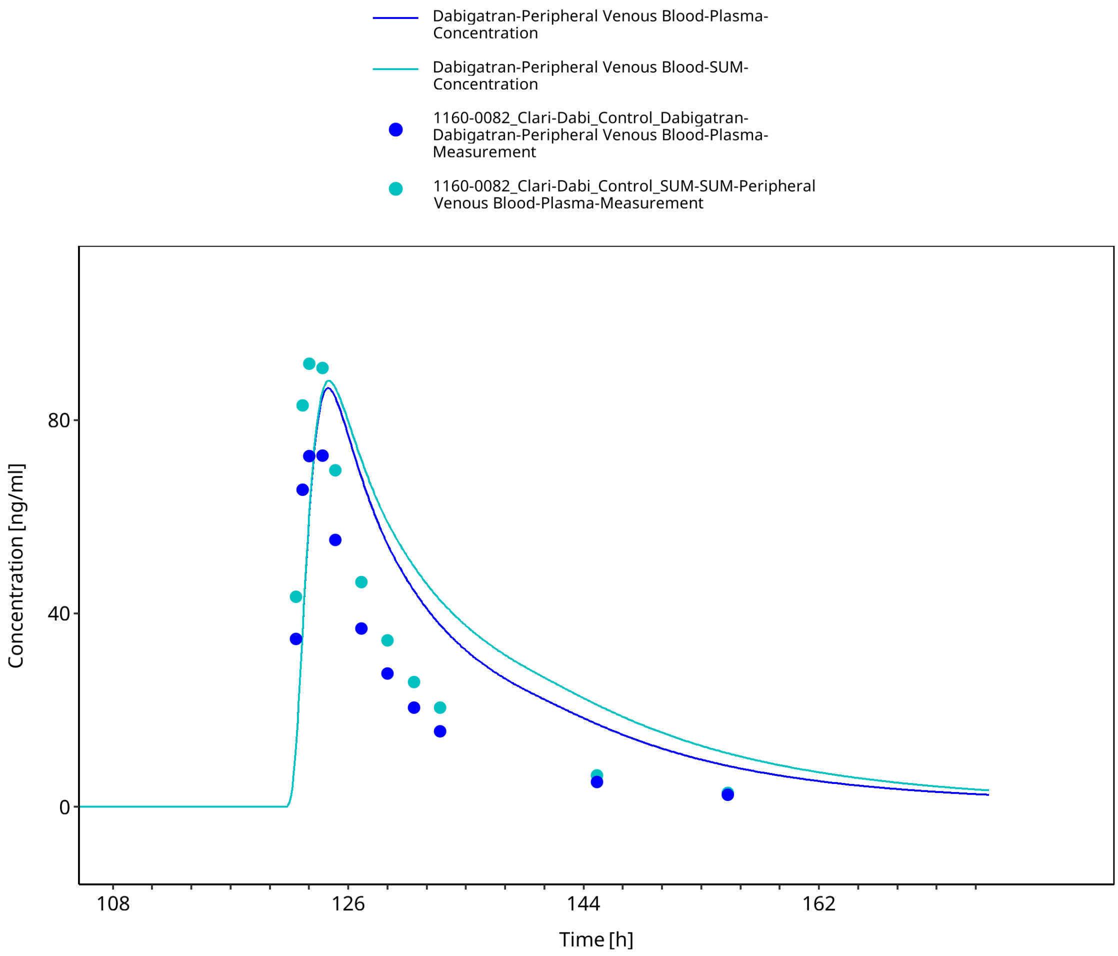

Figure 3-13: 1160-0082 - Dabigatran DDI Control - lin

4 Conclusion

The herein presented PBPK model adequately describes the pharmacokinetics of dabigatran etexilate, dabigatran, and dabigatran glucuronide in healthy adults.

In particular, it integrates the impact of Pgp on the absorption of dabigatran etexilate and was qualified against the clinical data of 9 different Pgp DDI studies for its application as Pgp substrate for DDI predictions (please see the DDI qualification report (https://github.com/Open-Systems-Pharmacology/COMPOUND-Model).

5 References

Blech 2008 Blech S, Ebner T, Ludwig-Schwellinger E, Stangier J, Roth W. The metabolism and disposition of the oral direct thrombin inhibitor, dabigatran, in humans. Drug Metab Dispos. 2008 Feb;36(2):386–399. doi: 10.1124/dmd.107.019083.

Delavenne 2013 Delavenne X, Ollier E, Basset T, Bertoletti L, Accassat S, Garcin A, Laporte S, Zufferey P, Mismetti P. A semi-mechanistic absorption model to evaluate drug–drug interaction with dabigatran: application with clarithromycin. Br J Clin Pharmacol. 2013 Jul;76(1):107–113. doi: 10.1111/bcp.12055.

Härtter 2012 Härtter S, Koenen-Bergmann M, Sharma A, Nehmiz G, Lemke U, Timmer W, Reilly PA. Decrease in the oral bioavailability of dabigatran etexilate after co-medication with rifampicin. Br J Clin Pharmacol. 2012 Sep;74(3):490–500. doi: 10.1111/j.1365-2125.2012.04218.x.

Kuepfer 2016 Kuepfer L, Niederalt C, Wendl T, Schlender JF, Willmann S, Lippert J, Block M, Eissing T, Teutonico D. Applied Concepts in PBPK Modeling: How to Build a PBPK/PD Model.CPT Pharmacometrics Syst Pharmacol. 2016 Oct;5(10):516-531. doi: 10.1002/psp4.12134.

Moj 2019 Moj D, Maas H, Schaeftlein A, Hanke N, Gómez-Mantilla JD, Lehr T. A Comprehensive Whole-Body PBPK Model of Dabigatran Etexilate, Dabigatran and Dabigatran Glucuronide in Healthy Adults and Renally Impaired Patients. Clin Pharmacokinet. 2019. doi: 10.1007/s40262-019-00776-y.

Paré 2013 Paré G, Eriksson N, Lehr T, Connolly S, Eikelboom J, Ezekowitz MD, Axelsson T, Haertter S, Oldgren J, Reilly P, Siegbahn A, Syvanen AC, Wadelius C, Wadelius M, Zimdahl-Gelling H, Yusuf S, Wallentin L. Genetic Determinants of Dabigatran Plasma Levels and Their Relation to Bleeding. Circulation. 2013 Apr;127(13):1404–12. doi: 10.1161/CIRCULATIONAHA.112.001233.

PK-Sim Ontogeny Database Version 7.3 (https://github.com/Open-Systems-Pharmacology/OSPSuite.Documentation/blob/38cf71b384cfc25cfa0ce4d2f3addfd32757e13b/PK-Sim%20Ontogeny%20Database%20Version%207.3.pdf)

PK-Sim Open Systems Pharmacology Suite PK-Sim® and MoBi® Documentation. (https://github.com/Open-Systems-Pharmacology/OSPSuite.Documentation)

Schlender 2016 Schlender JF, Meyer M, Thelen K, Krauss M, Willmann S, Eissing T, Jaehde U. Development of a Whole-Body Physiologically Based Pharmacokinetic Approach to Assess the Pharmacokinetics of Drugs in Elderly Individuals. Clin Pharmacokinet. 2016 Dec;55(12):1573-1589. doi: 10.1007/s40262-016-0422-3.

Zhao 2014 Zhao Y, Hu ZY. Physiologically based pharmacokinetic modelling and in vivo [I]/K(i) accurately predict P-glycoprotein-mediated drug-drug interactions with dabigatran etexilate. Br J Pharmacol. 2014 Feb;171(4):1043-53. doi: 10.1111/bph.12533.